Study Guide

From wholesale prices to consumer inflation: layered pass-through of global oil shocks in India

A plain-language walk-through for presentation and viva

Companion to the dissertation and journal paper by Aniket Pandey

Read this side by side with aniket-dissertation.docx and journal-paper.docx. Every result cited here comes from the model outputs in models/cpi and models/wpi.

How to use this guide

This guide is written for you, the student, and for anyone grading or listening to your defence. It is not a second dissertation. It is a companion. Read it in order the first time through. After that, jump to whichever section a specific question points to. Every Greek letter, every test, every model choice is explained in plain words before any formula is used.

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- Part 9: Results. Layer by layer with every number spelled out.
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Tip: *If a single question lands on one topic, turn to the Viva question bank in Part 12 first, then read the relevant methodology or results section for supporting detail.*

Part 1. The big picture in one page

1.1 What is the research about, in one line

We study how changes in the global price of crude oil reach different layers of prices in India, and we show that the effect is large near fuel prices but becomes much weaker by the time it hits the headline consumer price index.

1.2 Why does it matter

- India imports most of its crude oil, and crude is priced in US dollars. So every oil shock is also partly a rupee-dollar shock.
- Headline CPI is the number the Reserve Bank of India targets for inflation. If headline CPI moves only a little with oil, people may think oil no longer matters for Indian inflation. We show that is the wrong reading. Oil still matters. It matters most in fuel-related prices and becomes diluted by the time you reach the broad basket.
- For a policy reader this is useful. It says that headline CPI alone hides fuel-side pressure. You have to look at WPI Fuel and Power, retail petrol, and CPI Fuel and Light to see it.

1.3 The five price layers we look at

- Headline WPI: the broad wholesale price index.
- WPI Fuel and Power: the fuel-specific part of WPI.
- PPAC Delhi retail petrol: the price you actually pay at the pump.
- CPI Fuel and Light: the fuel-related part of the consumer basket.
- Headline CPI: the broad consumer price index.

1.4 The one-line punchline of the results

Positive cumulative pass-through CPT^{+} (a fancy name for 'how much of a one per cent oil rise shows up in the dependent price over the model's lag window') is about 0.52 for the preferred WPI Fuel and Power model, 0.35 for retail petrol, 0.18 for CPI Fuel and Light, 0.03 for headline WPI, and only 0.02 for headline CPI — and headline CPI is not even statistically significant. The numbers fall as we move from fuel to the broad consumer index. That fall is what we call layered attenuation.

1.5 One sentence on each technical ingredient

- Model: short-run asymmetric ADL in monthly log differences. ADL = autoregressive distributed lag. Asymmetric = positive and negative shocks enter separately. Short-run = we do not chase long-run relationships; the question is monthly response.
- Estimation: OLS, exactly the standard least squares you met in your first econometrics class, but with Newey-West HAC standard errors to fix the fact that monthly price data are auto-correlated and heteroscedastic.

- Inference: tests on whether the cumulative pass-through is zero, and whether positive and negative responses are equal (the symmetry test). Plus a restricted-residual circular block bootstrap with 4,999 replications as a robustness check.
- Diagnostics gate: a model only carries a main claim if it passes Breusch-Godfrey (no leftover serial correlation), HAC-RESET (functional form not obviously wrong), and recursive CUSUM (coefficients stable over time).

1.6 Research question in one sentence

How do global oil-price shocks transmit across India's wholesale, retail fuel, fuel-sensitive consumer, and headline consumer price layers, and where does this pass-through weaken?

1.7 Contribution in one sentence

We treat oil pass-through as a layered transmission problem across different price indices rather than as a single number linking crude to headline CPI, and we keep each layer's sample, shock variable, and diagnostics visible so nothing is hidden.

> **Remember:** *The dissertation does not claim oil causes Indian inflation. It claims oil shocks pass through strongly at fuel layers and weakly at the headline CPI layer, and the difference is statistically rejected by a Wald test.*

Part 2. Background: India, oil, WPI, and CPI

2.1 India and imported crude oil

India produces only a small share of the crude oil it consumes. Most of the crude used in Indian refineries is imported. The import bill is paid in US dollars because crude oil trades worldwide in dollars per barrel. That is why the rupee-dollar exchange rate is part of the story and not just a background variable.

When we talk about an ‘oil shock’ for India, we mean the movement in the rupee price of oil, $oil^{INR}_t = Brent^{USD}_t \times INR_t/USD_t$. If Brent rises or the rupee depreciates, this rupee oil price rises, and refiners and distributors face higher costs in rupees.

2.2 Why the dollar and the exchange rate matter

- A 10 per cent rise in Brent does not always lead to a 10 per cent rise in the rupee oil price. If the rupee appreciates at the same time, the rupee oil price rises by less than 10 per cent.
- A depreciation of the rupee, even without any Brent move, pushes up the rupee oil price.
- Studying the rupee oil price captures both channels at once. That is why our main WPI and headline CPI equations use the rupee oil shock as the explanatory variable.

2.3 The difference between WPI and CPI

Wholesale Price Index, WPI, measures prices at the wholesale or producer level. It is published by the Office of the Economic Adviser, which sits inside the Department for Promotion of Industry and Internal Trade, Ministry of Commerce and Industry. WPI has a fairly large share of mineral oils, fuel and power, and oil-intensive manufactured goods, which is why it is sensitive to oil shocks.

Consumer Price Index, CPI, measures prices at the household level. It is published by the Ministry of Statistics and Programme Implementation. The CPI basket is dominated by food, housing, services (like health and education), transport, and clothing. Fuel and Light is only one of several groups inside the CPI, so a pure fuel shock gets averaged down when you sum to the headline CPI.

Table 2.1. WPI and CPI at a glance.

Feature	WPI	CPI
Level of the economy measured	Wholesale / producer	Retail / household
Publisher	Office of the Economic Adviser, DPIIT, MoCI	Ministry of Statistics and Programme Implementation (MoSPI)
Base year used here	2011-12 = 100 (chained)	Processed series; official base periodically revised
Fuel and oil-intensive weight	High	Low (Fuel and Light is a sub-group)

Feature	WPI	CPI
Policy role	Useful for producer-side cost pressure	Main inflation target of the Reserve Bank of India

Note: Both indices are published monthly. The study chains successive WPI base years so a long monthly series is available from 1983 onward.

2.4 How fuel is priced in India, in short

Before 2010, retail petrol prices in India were administered. The government held them fixed for long stretches, and oil marketing companies (Indian Oil, Bharat Petroleum, Hindustan Petroleum) absorbed the difference through under-recoveries and fiscal support. So when Brent moved, retail petrol sometimes did not move at all for months.

In June 2010 petrol pricing was deregulated. The oil marketing companies were allowed to revise petrol prices more freely in line with international product prices. Diesel was deregulated in October 2014. The move to daily price revisions began in 2017. Under these rules, a Brent move now reaches retail petrol quickly, so the econometric pass-through we can measure from the data is bigger in the post-2010 sample than in the pre-2010 sample.

Careful wording: *We do not claim that deregulation caused the bigger post-2010 pass-through. Other things also changed: new CPI series in 2011, flexible inflation targeting in 2016, GST in 2017, tax-rate moves, and so on. The pre/post-2010 split is institutional evidence, not a clean experiment.*

2.5 Why a layered reading beats a single elasticity

Most earlier India papers pick one price index, usually headline WPI or headline CPI, and report a single number for how much oil passes through. That is fine, but it misses where the shock weakens. If you only look at headline CPI, you may get a small coefficient and think oil is unimportant. But retail petrol may be moving by a lot in the same data. Our layered design keeps all those layers in one frame so the reader can see the attenuation pattern.

2.6 The literature that anchors this work

- Mandal, Bhattacharyya and Bhoi (2012) studied oil price pass-through in India and found partial, asymmetric, and policy-sensitive transmission, especially through administered fuel prices.
- Bhanumurthy, Das and Bose (2012) analysed oil shock pass-through and its policy implications in India in an NIPFP working paper.
- Pradeep (2022) looked at diesel price reforms and asymmetric oil pass-through in India, using disaggregated data.
- Newey and West (1987) gave us the HAC standard errors we use for inference. The formula they introduced corrects standard errors for autocorrelation and heteroskedasticity.
- Bai and Perron (2003) gave the structural-break test we use as a background check for regime shifts.

- Kwiatkowski, Phillips, Schmidt and Shin (1992) gave the KPSS stationarity test we use alongside ADF and Phillips-Perron.

Part 3. Research question, objectives, and contribution

3.1 Main research question

How do global oil-price shocks transmit across India's wholesale, retail fuel, fuel-sensitive consumer, and headline consumer price layers, and where does this pass-through weaken?

Note what the question does not say. It does not ask ‘do oil prices cause Indian inflation’. It does not ask ‘what is the one true elasticity between Brent and headline CPI’. It asks how the shock moves across layers, and where it fades. This framing is the point of the whole paper.

3.2 Sub-questions you might be asked in the viva

- Does an oil shock visible in wholesale prices also show up in headline consumer inflation?
- Do positive and negative oil shocks have the same size of effect? This is the asymmetry question.
- Did fuel deregulation after 2010 change the size of pass-through? This is the pre/post-2010 question.
- Is headline CPI insulated from oil? If so, why? And if not, how much of the shock reaches it?

3.3 Research objectives

- Build a consistent monthly dataset of oil, exchange rate, WPI, retail petrol, and CPI layers.
- Estimate short-run asymmetric ADL models for each layer using OLS with Newey-West HAC standard errors.
- Summarise each layer's response with cumulative pass-through for positive shocks (CPT+) and negative shocks (CPT-).
- Test whether CPT+ differs from zero, whether CPT- differs from zero, and whether the two are equal (the symmetry test).
- Run diagnostics and robustness checks so the main claims are defensible.
- Present the results as an attenuation map with a formal Wald test on the consumer-side chain.

3.4 Scope and limits

- Monthly frequency only. We do not use daily or quarterly data.
- Short-run pass-through only. We do not estimate long-run equilibrium relationships, so there is no NARDL bounds test, no error correction model, and no cointegration claim.
- India as a whole. Retail petrol prices are taken from Delhi and treated as a retail-channel proxy, not a national retail measure.
- Reduced-form models. We estimate projections, not structural causal effects.

3.5 Contribution

The main contribution is the layered framing. Instead of one regression of headline CPI on oil, we fit five reduced-form equations, one per layer, and put the cumulative pass-through numbers side by side. This makes the attenuation pattern visible and testable with a formal Wald test on the common-sample consumer-side chain.

A second smaller contribution is methodological hygiene. We use a strict diagnostic gate (Breusch-Godfrey, HAC-RESET, recursive CUSUM) and we report models that fail the gate as context rather than as claim-bearing estimates. For example, the full-sample pooled WPI Fuel and Power model fails HAC-RESET, so we report it as context and move the wholesale fuel claim to the preferred post-2010 excluding-COVID model which clears the gate cleanly.

3.6 What we deliberately did not do

- No long-run cointegration analysis. Our claim is about monthly transmission, so long-run tools would be overkill and potentially misleading.
- No structural VAR decomposition. We report predictive precedence from Granger tests but we do not claim a structural shock identification.
- No disaggregated CPI sub-component regressions beyond Fuel and Light. That is explicitly listed as a future extension.

One-line contribution: *Oil pass-through in India is best read as a layered attenuation map, not as a single elasticity, and we provide the numbers and the formal test that support that reading.*

Part 4. Data: what is it, where does it come from, and why

4.1 All series at monthly frequency

Every series we use is monthly. Different series start in different years because the underlying publications began at different dates. This is why the sample windows are not the same across layers.

4.2 Crude oil price: Brent, from the World Bank Pink Sheet

‘Brent crude’ is a benchmark crude oil produced in the North Sea. ‘Brent price’ is the price of a barrel of Brent crude in US dollars. World traders use Brent as a standard, so it is the global oil price most economists use for applied work on India.

We take the monthly Brent price from the World Bank Commodity Price Data, the publication commonly called the Pink Sheet. It is a monthly bulletin published by the World Bank with prices of major energy, metals, and agricultural commodities. The WPI model also cross-checks against FRED series POILBREUSDM. FRED stands for Federal Reserve Economic Data, maintained by the Federal Reserve Bank of St. Louis.

4.3 Exchange rate: INR per USD, from FRED

INR per USD is how many rupees you pay for one dollar. It goes up when the rupee depreciates. We take the monthly average INR/USD rate from the FRED series EXINUS (the CPI pipeline uses EXINUS; the WPI pipeline uses the updated EXINUS_latest file). FRED pulls this from the Federal Reserve's H.10 statistical release.

4.4 Rupee oil price: constructed, not downloaded

The rupee oil price, oil^{INR}_t, is built as Brent in dollars multiplied by the INR/USD rate: $oil^{INR}_t = Brent^{USD}_t \times INR_t/USD_t$. This is the main shock variable for the WPI and headline CPI equations.

4.5 Wholesale Price Index (WPI): from the Office of the Economic Adviser

The Office of the Economic Adviser (OEA) inside the Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry, Government of India, publishes the WPI every month. We use two WPI series: headline WPI (the whole index) and WPI Fuel and Power (the sub-index that tracks fuel, power, and mineral oil items in the wholesale basket).

The OEA has published WPI under several base years, 1981-82, 1993-94, 2004-05, and now 2011-12 = 100. To get a continuous monthly series from May 1983 to March 2026, we chain the older vintages using the OEA's official chain factors and rebase everything to 2011-12 = 100. The pipeline verifies splices at the overlap dates so that we do not introduce visible level breaks.

4.6 PPAC Delhi retail petrol: from the Petroleum Planning and Analysis Cell

PPAC, the Petroleum Planning and Analysis Cell, is a unit under the Ministry of Petroleum and Natural Gas. It publishes a ready reckoner with monthly retail prices of petrol and diesel in major Indian cities. We use the Delhi retail petrol price from August 2004 to December 2024 as our direct retail-fuel layer. We choose Delhi because it has one of the longest, most consistent monthly series.

4.7 Consumer Price Index (CPI): from MoSPI

MoSPI, the Ministry of Statistics and Programme Implementation, publishes the all-India CPI every month. We use two CPI series: headline CPI (the whole index) and CPI Fuel and Light (the sub-group that collects household energy items like LPG, kerosene, and electricity-related charges).

In practice, headline CPI is loaded from a processed file inside the CPI pipeline (the FRED OECD series INDCPIALLMINMEI is the headline source; MoSPI is the authority). CPI Fuel and Light is the processed MoSPI all-India component series used starting in May 2011, when the current harmonised series is continuously available.

4.8 Index of Industrial Production (IIP): MoSPI

IIP is a monthly index of manufacturing and industrial output published by MoSPI. We use its monthly growth rate as a control for domestic activity or demand in the retail-petrol and headline CPI equations. The idea is that some inflation movement is caused by demand conditions, not by oil, and we do not want the oil coefficient to pick up that other story.

4.9 Summary table of series and their roles

Table 4.1. Every series used in the paper, with its full source name, transformation, active sample, and role.

Series	Source (full name)	Transformation	Active sample	Role in the paper
Brent crude price	World Bank Commodity Price Data (Pink Sheet)	Monthly log difference	Matched to layer	Global oil shock
INR/USD exchange rate	Federal Reserve Economic Data (FRED EXINUS / EXINUS_latest)	Monthly log difference	Matched to layer	Exchange-rate component
Rupee oil price	Brent multiplied by INR/USD	Monthly log difference	Matched to layer	Domestic oil shock
Headline WPI	Office of the Economic Adviser (DPIIT, MoCI), 2011-12 = 100	Monthly log difference	1983-05 to 2026-03	Wholesale endpoint
WPI Fuel and Power	Office of the Economic Adviser (DPIIT, MoCI), 2011-12 = 100	Monthly log difference	1995-05 to 2026-03	Wholesale fuel layer

Series	Source (full name)	Transformation	Active sample	Role in the paper
PPAC Delhi retail petrol	Petroleum Planning and Analysis Cell (MoPNG) ready reckoner	Monthly log difference	2004-08 to 2024-12	Retail fuel layer
CPI Fuel and Light	Ministry of Statistics and Programme Implementation (MoSPI)	Monthly log difference	2011-05 to 2024-12	Consumer fuel bridge
Headline CPI	MoSPI, via processed CPI source (OECD INDCPIALLMINMEI / FRED)	Monthly log difference	2004-08 to 2024-12	Consumer endpoint
Index of Industrial Production	MoSPI	Monthly log difference	Matched to layer	Activity control

Note: Samples differ across layers because the official series became available in different years. We deliberately did not force a common sample; instead we run a common-sample CPI-chain check separately (see Part 9).

4.10 Why samples are different, and why that is fine

- Headline WPI has the longest window (1983 onward) because the OEA series goes back that far. We use it fully, since more data gives better inference.
- WPI Fuel and Power starts in 1995 because that is when the OEA component series is continuously available in a chainable form.
- PPAC retail petrol starts in August 2004 because that is the earliest consistent monthly series from PPAC.
- Headline CPI uses the same post-2004 window so the retail petrol and headline CPI are on the same 20-year consumer-side window.
- CPI Fuel and Light starts in May 2011 because that is when the current harmonised MoSPI series begins. We treat it as bridge evidence for this reason, not as the main headline model.

Viva tip: *If a teacher asks why the sample windows differ, answer: 'Each series is used from the earliest date its official publisher makes it available. Headline WPI goes back to 1983. CPI Fuel and Light starts in 2011, which is why we treat it as bridge evidence and not as our main consumer-side claim.'*

Part 5. Variables: log differences, shocks, and the rupee oil price

5.1 Why we use log differences and not raw levels

Prices grow roughly exponentially over time. If we regress the level of CPI on the level of Brent, we get a misleading 't-statistic' because both variables trend together, even when no real relationship exists. This is the classic spurious regression problem from Granger and Newbold (1974). To avoid it, we difference the logs.

Logarithms turn a proportional change into a level change. If a price goes up by one per cent, the log goes up by about 0.01. Taking the difference of the log is therefore a clean way to measure monthly growth in per cent terms.

5.2 The log-difference formula

$$\Delta x_t = 100 \times \left(\ln\left(x_t\right) - \ln\left(x_{t-1}\right)\right) \qquad (5.1)$$

Here x_t is the value of a series in month t and x_{t-1} is its value the previous month. The factor of 100 turns the number into a percentage. So Δx_t is the percentage growth of x from month t−1 to month t.

Intuition: *If Brent goes from 100 dollars in one month to 101 dollars the next, then $\Delta(\text{Brent})_t$ is approximately 100 × (ln 101 − ln 100) ≈ 0.995, very close to 1. We read this as about a one per cent rise.*

5.3 Splitting shocks into positive and negative parts

For every shock variable we build two 'twins': one that is zero whenever the shock is negative, and one that is zero whenever the shock is positive. Formally, $\Delta x^+_t = \max(\Delta x_t, 0)$ and $\Delta x^-_t = \min(\Delta x_t, 0)$. The first is non-negative. The second is non-positive. Added together they reconstruct the original shock.

Why do this split? Because an oil-price rise and an oil-price fall may pass through at different speeds. Retailers often raise fuel prices quickly and cut them slowly, a pattern the pass-through literature calls 'rockets and feathers'. The split lets the data test whether CPT^+ and CPT^- are equal without forcing them to be.

5.4 Cumulative pass-through, CPT+ and CPT−

In the ADL model we put several lags of Δx^+_t and several lags of Δx^-_t on the right-hand side. Each lag has its own coefficient β^+_0, β^+_1, … and β^-_0, β^-_1, …. The cumulative pass-through for positive shocks is simply the sum of those β^+ coefficients over the lag window. $CPT^+ = \Sigma_{j=0}^{q} \beta^+_j$. Likewise $CPT^- = \Sigma_{j=0}^{q} \beta^-_j$.

$CPT^{+} = 0.3$ means: a one per cent positive oil shock this month is associated with, in total across this month and the following q months, about a 0.3 per cent change in the dependent price index. CPT^{-} is the twin number for negative shocks.

Careful: *Since Δx^{-} is non-positive by construction, CPT^{-} times a negative shock value produces a negative price response on average. Do not read CPT^{-} as if it were a separate positive shock.*

5.5 Why we multiply the log difference by 100

The factor of 100 is cosmetic. It simply turns log differences into percentages so that every coefficient reads as 'percentage response per one per cent shock'. Without the 100, the coefficients would be the same story told in decimals (0.003 instead of 0.30).

5.6 Rupee oil price step by step

We construct the rupee oil price in two steps:

```
# Step 1: align dates and pick monthly series
brent_usd       <- load_world_bank_pink_sheet()
inr_per_usd     <- load_fred_EXINUS()

# Step 2: multiply them at each date to get rupee oil
oil_inr[t]      <- brent_usd[t] * inr_per_usd[t]

# Step 3: make it a monthly change
dln_oil_inr[t]  <- 100 * ( log(oil_inr[t]) - log(oil_inr[t-1]) )

# Step 4: split into positive and negative shock
dln_oil_inr_pos[t] <- max(dln_oil_inr[t], 0)
dln_oil_inr_neg[t] <- min(dln_oil_inr[t], 0)
```

The same step-by-step logic applies to Brent alone (for the retail petrol equation) and to PPAC petrol (for the CPI Fuel and Light bridge equation). The only difference is which input series plays the role of 'x'.

5.7 Controls you will see in the equations

- ΔIIP_t: the monthly growth of the Index of Industrial Production. A demand-side control for the retail petrol and headline CPI equations.
- D_petrol: a dummy variable that is 0 before June 2010 and 1 after that, to allow the intercept to shift after petrol deregulation.
- D_diesel: a dummy variable that is 0 before October 2014 and 1 after, to mark diesel deregulation.
- D_covid: a dummy that flags the unusual months during COVID (April-September 2020 in the WPI models; April 2020 in the CPI models).
- μ_m: eleven monthly seasonal dummies (one month is the reference), to soak up residual seasonality after month-on-month differencing.

5.8 Why the shock variable differs across layers

We use the rupee oil price as the shock for WPI, WPI Fuel and Power, and headline CPI, because Indian producers and the economy as a whole care about the rupee cost of imported crude, which bundles Brent and the exchange rate.

For retail petrol we use Brent alone, because retail petrol prices are pegged (under market-linked pricing) to international product prices that are very close to the dollar benchmark. Tax and pricing rules absorb much of the exchange-rate component separately.

For CPI Fuel and Light we use PPAC retail petrol as the shock, because the question here is specifically how retail fuel movements enter the fuel-sensitive consumer basket. Using oil directly would mix the Brent-to-petrol stage and the petrol-to-CPI-Fuel-and-Light stage.

Part 6. Methodology: the ADL model in plain language

6.1 What is an ADL model

ADL stands for Autoregressive Distributed Lag. Let us unpack that:

- ‘Autoregressive’ means the dependent variable depends on its own past values. If last month's inflation in WPI Fuel and Power was high, this month's is also likely to be a bit higher on average. So we put lags of the dependent variable on the right-hand side.
- ‘Distributed lag’ means the explanatory variable enters not only at time t, but also at t−1, t−2, and so on. We let the effect of an oil shock spread out over several months, because in reality refiners, taxes, and supply chains take time to pass costs on.
- ‘Short-run asymmetric’ means (a) we are interested in the monthly dynamics, not the long-run level relationship, and (b) positive and negative shocks are entered as separate variables.

6.2 The full ADL equation with every term explained

$$\Delta y_t = \alpha + \sum_{i=1}^{p} \varphi_i \Delta y_{t-i} + \sum_{j=0}^{q} \beta_j^{+} \Delta x_{t-j}^{+} + \sum_{j=0}^{q} \beta_j^{-} \Delta x_{t-j}^{-} + \gamma' Z_t + \mu_m + \varepsilon_t \quad (6.1)$$

6.3 Every symbol, one by one

Table 6.1. Every Greek letter and symbol in the ADL equation, translated.

Symbol	Name	What it means in plain language
Δy_t	Delta y sub t	The percentage monthly change in the dependent price index at month t. E.g. this month's inflation in headline CPI.
α	Alpha	The intercept, or baseline monthly inflation when all right-hand-side variables are zero. It sets the ‘average level’ of the equation.
φ_i	Phi sub i	The coefficient on the i-th lag of the dependent variable. Measures how persistent inflation is in that index.
p	p	The number of own-lags used. For WPI models we use p = 12 to capture the annual cycle; for CPI and retail petrol we use p = 3, chosen by AIC over p = 1 to 4.
β^{+}_{j}	beta plus sub j	The coefficient on the j-th lag of the positive oil shock. Its sign tells us how a positive oil shock j months ago affects today's inflation.
β^{-}_{j}	beta minus sub j	The coefficient on the j-th lag of the negative oil shock.
q	q	The number of shock lags. For WPI we use q = 6 (wholesale channels are slower). For retail petrol and CPI we use q = 3 (fuel pricing and consumer baskets respond faster).
Z_t	Z sub t	A vector of extra controls. For the retail petrol equation it contains IIP growth and the petrol/diesel deregulation dummies. For headline CPI it contains the same, plus a COVID dummy.
γ	Gamma	The vector of coefficients on those controls. The prime (γ') just means ‘multiply γ against the vector Z_t row by row’.

Symbol	Name	What it means in plain language
μ_m	Mu sub m	The month-of-year dummy. Eleven dummies for eleven months, with one month as the reference. Soaks up residual seasonality that survives log-differencing.
ε_t	Epsilon sub t	The error term. Everything the model did not explain. It is allowed to be heteroscedastic and autocorrelated; that is why we fix the standard errors later.

Note: Σ is the sum symbol. The subscript below Σ is where the index starts; the superscript above Σ is where it ends.

6.4 How the coefficients α, β, γ, φ, μ are derived: OLS in plain terms

All the coefficients in the ADL equation are estimated by ordinary least squares, OLS, exactly the same OLS you met in first-year econometrics. Nothing exotic.

OLS picks the numbers $\hat{\alpha}$, $\hat{\beta}^{+}_{j}$, $\hat{\beta}^{-}_{j}$, $\hat{\gamma}$, $\hat{\varphi}_i$, $\hat{\mu}_m$ that make the sum of squared residuals $\Sigma\ \hat{\varepsilon}^2_t$ as small as possible. A residual $\hat{\varepsilon}_t$ is the gap between the actual Δy_t we observe and the value our equation would predict for that month. OLS says: choose the coefficients so these gaps, squared and summed over all months in the sample, are minimised.

> **Why squares, not absolute values:** *Squaring punishes big misses more than small ones and makes the math tractable. Under classical assumptions it is also the best-linear-unbiased estimator by the Gauss-Markov theorem.*

6.5 A tiny worked example of OLS

Imagine a toy regression: $\Delta y_t = \alpha + \beta\ \Delta Brent_t + \varepsilon_t$ with just three observations:

Table 6.2. Toy data for the OLS worked example.

Month	ΔBrent	Δy
t=1	1.0	0.4
t=2	2.0	0.9
t=3	3.0	1.3

If we choose α = 0 and β = 0.45, the predicted Δy values are 0.45, 0.90, 1.35. The residuals are (0.4−0.45)=−0.05, (0.9−0.90)=0.0, (1.3−1.35)=−0.05. The sum of squared residuals is 0.005. OLS tries every possible pair (α, β) and picks the pair with the smallest sum of squared residuals. In practice the statistical package solves a simple matrix equation to find this minimum in one step.

The same logic scales up to our ADL equation with many lags and many controls. Instead of two coefficients the software estimates thirty or more, but the principle is identical: the chosen coefficients are the ones that minimise the total squared prediction error.

```
# Pseudo-code for OLS fit used inside the R scripts
y  <- dln_cpi                              # dependent variable
X  <- cbind(1,                             # intercept
            dln_cpi_lag1, dln_cpi_lag2, dln_cpi_lag3,
            dln_oil_inr_pos, dln_oil_inr_pos_lag1,
            dln_oil_inr_pos_lag2, dln_oil_inr_pos_lag3,
```

```
             dln_oil_inr_neg, dln_oil_inr_neg_lag1,
             dln_oil_inr_neg_lag2, dln_oil_inr_neg_lag3,
             dln_iip, D_petrol, D_diesel, D_covid,
             month_dummies)

beta_hat <- solve(t(X) %*% X) %*% (t(X) %*% y)    # OLS closed-form
residuals <- y - X %*% beta_hat
```

6.6 How many lags to use: the (p, q) in ADL(p, q)

ADL(p, q) is shorthand for 'p lags of the dependent variable and q lags of the shock variable'. The choices we use come from the lag-selection tables inside the model pipelines, not from guesswork.

- For the headline WPI and WPI Fuel and Power equations we use ADL(12, 6). Twelve own-lags capture the annual cycle in wholesale inflation even after month dummies. Six shock-lags let the oil effect stretch half a year, which matches how wholesale passes from refineries through the supply chain.
- For the CPI Fuel and Light bridge and the headline CPI equations we use ADL(3, 3). Three own-lags because AIC over p in {1, 2, 3, 4} prefers three. Three shock-lags because the retail fuel and consumer-basket transmission is fast and deeper lags add noise.
- For the PPAC retail petrol equation we also use ADL(3, 3). Retail petrol adjusts quickly under post-2010 rules, so longer lags are not needed.

AIC stands for Akaike Information Criterion. It rewards models that fit the data well and penalises models with too many parameters. Lower AIC is better. BIC (Bayesian Information Criterion) is the same idea with a stronger penalty for extra parameters. HQIC (Hannan-Quinn Information Criterion) is similar. We pick the lag order that minimises AIC among candidate orders, and confirm with BIC and HQIC when possible.

6.7 Why we include month dummies

Some months are systematically more inflationary than others, for reasons unrelated to oil: the monsoon hits food prices, festival seasons shift consumption patterns, tax-revision calendars align with fiscal years. Taking monthly log differences removes part of this seasonality but not all. Adding eleven monthly dummies for February through December (with January as the omitted reference month) absorbs the rest.

6.8 Why we include policy dummies and a COVID dummy

- D_petrol captures the intercept shift after June 2010 petrol deregulation. Without it, the ADL equation would average pre- and post-2010 behaviour together and could bias the shock coefficients.
- D_diesel does the same for October 2014 diesel deregulation.

- D_covid flags the unusual April-September 2020 period. During COVID, oil prices collapsed and Indian retail prices did not move the way the usual pricing rules would suggest, because demand was extraordinary. Leaving those months in without a flag can distort the estimates.

Two ways we handle COVID: *For most models we include a D_covid dummy. For the preferred WPI Fuel and Power model we go further and drop April-September 2020 entirely from the sample, because the functional-form diagnostic test otherwise fails due to these outliers.*

6.9 One equation, five estimations

Table 6.3. The five estimated equations, their shock and dependent variables, lag order, and controls.

Layer	Δy (dependent)	Δx (shock)	ADL(p,q)	Main controls in Z
Headline WPI	WPI inflation	Rupee oil shock	ADL(12,6)	Month fixed effects
WPI Fuel and Power (preferred)	WPI Fuel and Power inflation	Rupee oil shock	ADL(12,6)	Exchange-rate controls, month dummies, post-2010 sample, excludes Apr-Sep 2020
PPAC retail petrol	Delhi retail petrol inflation	Brent shock	ADL(3,3)	IIP growth, D_petrol, D_diesel, D_covid, month dummies
CPI Fuel and Light (bridge)	CPI Fuel and Light inflation	PPAC petrol shock	ADL(3,3)	Month dummies
Headline CPI (M1)	Headline CPI inflation	Rupee oil shock	ADL(3,3)	IIP growth, D_petrol, D_diesel, D_covid, month dummies

Note: Every row is a separate OLS regression. The paper does not combine them into one simultaneous system.

Part 7. Inference: p-values, HAC, Wald, bootstrap, Granger

7.1 What is a p-value in one line

A p-value is the probability of seeing a coefficient at least as extreme as the one we estimated, if the true coefficient were zero. A small p (conventionally below 0.05) tells us the estimate is unlikely to be a random accident, so we say the coefficient is 'statistically significant' at that level.

Simple translation: *Coefficient = 0.30 and p = 0.002 means: 'The data say the effect is about 0.30, and there is only a 0.2 per cent chance we would see something this extreme if the real effect were zero. So we believe the effect is real.'*

7.2 The standard error and why monthly data need HAC

Every coefficient has a standard error, which measures how noisy that coefficient estimate is. The p-value is the coefficient divided by its standard error, compared to a reference distribution. If the standard error is wrong, the p-value is wrong.

Classical OLS standard errors assume residuals are independent and have the same variance ('no heteroskedasticity, no autocorrelation'). Monthly inflation data break both assumptions. Errors are often correlated across months, and their variance changes with regimes like COVID. So we use Newey-West HAC standard errors instead.

- HAC = Heteroskedasticity and Autocorrelation Consistent.
- Named after Newey and West (1987), who derived them.
- They adjust the covariance matrix of the coefficients so the p-values remain valid even when residuals are heteroskedastic and autocorrelated.
- The adjustment uses a kernel (we use the Bartlett kernel) and a bandwidth; we use a data-dependent bandwidth with floor($0.75 \times N^{1/3}$) as the rule for the main wholesale models.

7.3 The three hypotheses we test on cumulative pass-through

For every layer we report three tests. First, H_0: $CPT^+ = 0$. Does the positive-shock response sum to zero, or is there a real positive response? Second, H_0: $CPT^- = 0$. Same question for negative shocks. Third, H_0: $CPT^+ = CPT^-$. This is the symmetry test. Can we treat positive and negative shocks as equivalent in size?

All three tests are linear restrictions on combinations of the β coefficients. They are standard Wald tests. A Wald test checks whether the coefficients estimated from the data are far from the values in the null hypothesis, in a way that is weighted by the HAC covariance matrix.

7.4 How a Wald test works, intuitively

Imagine we estimated $CPT^+ = 0.35$ with a standard error of 0.08. The null hypothesis $CPT^+ = 0$ says the true CPT^+ is zero. Our estimate is $0.35 / 0.08 \approx 4.4$ standard errors away from zero. A test statistic that big gives a very small p-value, so we reject the null and conclude CPT^+ is not zero.

Similarly, the symmetry test looks at ($CPT^+ - CPT^-$) and its standard error under the HAC covariance. If the difference is small relative to its standard error, we do not reject symmetry.

7.5 The bootstrap symmetry check

The Wald test relies on large-sample theory. In finite samples the p-value can be slightly off. A bootstrap re-computes the p-value without relying on asymptotic theory. We use a restricted-residual circular block bootstrap with 4,999 replications.

- 'Restricted-residual' means we enforce the null ($CPT^+ = CPT^-$) when we resample, so that the distribution we build is the one under the null.
- 'Circular block' means we resample residuals in blocks to preserve the within-block autocorrelation that monthly data exhibit. The 'circular' part wraps around the end of the sample so every point has the same chance of being included.
- We do this 4,999 times, each time computing a fake test statistic. The bootstrap p-value is the fraction of fake statistics at least as extreme as the one we saw in the real data.

Why 4,999: *Odd numbers avoid ambiguity when computing the fraction. 4,999 is a conventional, stable choice; the results change very little if you use 9,999 instead.*

7.6 Granger causality: predictive precedence, not real causality

A Granger test asks: if we already know current and past values of x and y, does adding past values of oil improve our forecast of y? If yes, oil 'Granger-causes' y. That is a predictive statement, not a structural causal claim. Oil can Granger-cause CPI without being the sole cause of CPI movements.

We report these tests alongside the ADL results to check whether the direction of predictive information is consistent with our layered reading. For example, rupee oil Granger-causes headline WPI and WPI Fuel and Power, while the reverse directions do not hold. This matches the idea that oil leads wholesale, not the other way around.

What to say in the viva: *If asked 'does oil cause inflation', answer: 'We test predictive precedence, not structural causality. Oil Granger-causes wholesale inflation in our sample, but we do not claim it is the only cause.'*

7.7 Reading the main numbers in our paper

When we report $CPT^+ = 0.3459$ ($p < 0.001$) for retail petrol, we mean: our point estimate of the cumulative response to a positive one per cent Brent shock, summed over three shock lags, is about 0.35 per cent, and under the HAC standard errors the probability of seeing something that big under the null $CPT^{+} = 0$ is below 0.001. That is very strong evidence that the positive cumulative pass-through is real.

Asymmetry $p = 0.0999$ for retail petrol means: the HAC-based Wald test for H_0: $CPT^{+} = CPT^{-}$ gives a p-value of about 0.10. This is borderline. We say it rejects symmetry only at the 10 per cent level and describe it as suggestive, not decisive.

Part 8. Every test, in plain language

This part covers every statistical test used anywhere in the paper. For each test you will see four things: what problem it checks, what its null hypothesis H0 says in plain words, a one-sentence rule for reading the p-value, and a quick memory hook. Read the memory hook aloud once and you will remember the test on the first try.

8.1 The memory rule that works for every test

Every test gives one number, the p-value. There is only one rule you need: if $p < 0.05$, we reject H_0. If $p \geq 0.05$, we fail to reject H_0. 'Reject' means 'the data disagree with H_0'. 'Fail to reject' means 'the data are fine with H_0'. Whether rejecting is good news or bad news depends on what the test is checking. That is where students go wrong, so each test below tells you which direction you want.

Rule of thumb: *Small p (below 0.05) = reject H0. Large p (0.05 or more) = fail to reject H0. Whether that is good or bad depends on what H0 says.*

8.2 Unit root tests: is the series stationary?

A 'stationary' series has a stable mean and variance over time. A 'unit root' series drifts indefinitely and needs to be differenced to calm it down. If we run a regression on two unit-root series in levels, we get a spurious 'significant' result just because both are trending. So we test for unit roots before fitting the model.

ADF test (Augmented Dickey-Fuller)

- What it checks: does the series have a unit root?
- H0 (null): the series has a unit root (non-stationary).
- Rule: small p = reject H0 = series is stationary (good for us).
- Memory hook: ADF says 'You are non-stationary until proven stationary.' A low p breaks the default verdict.

PP test (Phillips-Perron)

- What it checks: same question as ADF, but it handles autocorrelation in a non-parametric way.
- H0: the series has a unit root.
- Rule: small p = reject = stationary.
- Memory hook: PP is ADF's cousin with a different cleaning routine.

KPSS test (Kwiatkowski-Phillips-Schmidt-Shin, 1992)

- What it checks: the OPPOSITE of ADF. Starts from stationarity.
- H0: the series is stationary.
- Rule: small p = reject = NOT stationary (has a unit root).

- Memory hook: KPSS is the ‘flip’. If KPSS says $p < 0.05$, that is BAD news for stationarity. Read it backward compared with ADF.

Why we use all three: *ADF/PP and KPSS have opposite nulls. If ADF rejects (p small) AND KPSS fails to reject (p large), both say the same thing: the series is stationary. If they disagree, we read it as ambiguous and look at more evidence.*

Zivot-Andrews test

- What it checks: is there a unit root when the data might have ONE structural break at an unknown date?
- H0: unit root, possibly with a break.
- Rule: small p = reject = stationary with a break.
- Memory hook: ZA is ADF ‘with a crack line’. It allows one sudden jump without being fooled by it.

In our paper: all the price series in levels have unit roots. After taking log differences they become stationary by ADF, PP, and KPSS, with a small KPSS caveat for headline WPI inflation. That caveat is handled by the Bai-Perron break test, not by differencing again.

8.3 Structural break tests: did the relationship shift in time?

Bai-Perron (2003) test

- What it checks: are there one or more unknown break points where the relationship shifts?
- H0: no break, parameters are constant.
- Rule: small p = reject = at least one break exists.
- Memory hook: Bai-Perron is a ‘break detective’. It scans the whole sample and says ‘a shift happened around month X’.

In our paper: where Bai-Perron detects breaks in headline WPI inflation, we handle them with the institutional pre/post-2010 split rather than adding dummies for every mathematical break. This keeps the analysis anchored to fuel-pricing policy.

8.4 Residual diagnostics: is the fitted model well-behaved?

After fitting the ADL, the residuals $\hat{\varepsilon}_t$ should look like clean noise. These tests check whether they do. Anything unusual in the residuals means the model is missing something.

Breusch-Godfrey test (BG), the leftover-pattern test

- What it checks: do the residuals still have autocorrelation? In other words, is last month's error predicting this month's error?
- H0: no serial correlation in residuals.
- Rule: LARGE p is what we want. Large p = fail to reject = residuals are clean. Small p = bad, residuals still have pattern.

- Memory hook: BG scans the residuals for leftover patterns. Pass it like a quality-control inspection at the end of a factory line.

We run BG at 12 lags because the data are monthly and annual seasonality could sneak in. Our claim-bearing models all pass BG ($p > 0.05$).

Breusch-Pagan test (BP), the variance-stability test

- What it checks: is the residual variance constant, or does it grow and shrink with the size of the right-hand-side variables?
- H0: constant variance (homoskedasticity).
- Rule: large p = residuals have stable variance = happy.
- Memory hook: BP listens to whether the noise is 'loud sometimes, quiet sometimes'. If yes, the standard errors need fixing. We use HAC errors either way, so BP is a health check, not a gate.

Jarque-Bera test (JB), the normality test

- What it checks: do the residuals follow a bell-shaped (normal) distribution?
- H0: residuals are normally distributed.
- Rule: large p = residuals look normal.
- Memory hook: JB is a 'bell-shape score'. Failing it does not kill OLS (we still get consistent estimates), but very large fails flag outliers worth investigating.

Ramsey RESET test (HAC-RESET), the missing-shape test

- What it checks: did we use the wrong functional form? For example, is there a nonlinear effect the linear ADL missed?
- How it works, intuitively: RESET takes our fitted $\hat{y}$, squares and cubes it, and sticks $\hat{y}^2$ and $\hat{y}^3$ into the regression as extra variables. If those extra terms are jointly significant, then our original straight-line model was missing curvature.
- H0: the functional form is correct.
- Rule: large p = 'straight line is fine, no missing curvature'. Small p = 'the model is missing nonlinearity' = bad.
- Memory hook: RESET adds two 'bendy' regressors to the equation. If they matter, the original shape was wrong.

We run RESET under the HAC covariance so it stays valid with autocorrelated, heteroskedastic residuals. That is what 'HAC-RESET' means.

Recursive CUSUM test, the stability test

- What it checks: do the coefficients stay roughly constant as we add data one month at a time, or do they drift?
- How it works, intuitively: fit the model on the first few months. Predict the next month. Record the forecast error. Do it again with one more month of data. Plot the cumulative sum of these errors.
- H0: coefficients are stable over time.

- Rule: a large p (and a CUSUM line that stays inside its 5 per cent confidence band) = stable coefficients. A small p = instability somewhere.
- Memory hook: CUSUM is a ‘running tab’ of forecast errors. If the tab ever runs too far from zero, the model's coefficients are shifting under our feet.

OLS CUSUM test

- Same idea as recursive CUSUM but built from the residuals of a single full-sample OLS fit rather than a growing-window fit.
- Useful as a cross-check. We report recursive CUSUM as the main gate test because it is more sensitive to drift.

8.5 The mandatory model gate, in one mental picture

Before a model can carry a main claim in our paper it must pass three doors in a row. Think of it as passport control.

- Door 1: BG12 (Breusch-Godfrey at 12 lags). ‘Are the residuals free of leftover patterns?’
- Door 2: HAC-RESET. ‘Is the functional form right?’
- Door 3: Recursive CUSUM. ‘Are the coefficients stable over time?’

Any model that fails even one door is downgraded to ‘context’ or dropped from the main text. No exceptions. This rule is set BEFORE looking at the coefficients, so we cannot cherry-pick.

Table 8.1. Gate result for every specification in the paper.

Specification	BG12	HAC-RESET	Rec-CUSUM	Result
Headline WPI (1983-2026)	Pass	Pass	Pass	Main WPI claim
WPI Fuel and Power, pooled full sample	Pass	FAIL	Pass	Context only
WPI Fuel and Power, post-2010, excluding Apr-Sep 2020	Pass	Pass	Pass	Preferred wholesale fuel claim
Headline CPI M1 (2004-2024)	Pass	Pass	Pass	Main CPI endpoint
PPAC retail petrol (2004-2024)	Pass	Pass	Pass	Mechanism accepted
CPI M2, M3 alternatives	Varies	FAIL	Varies	Not claim-bearing

8.6 How we made WPI Fuel and Power pass HAC-RESET

This is the most-asked methodology question the teacher can drop on you. Read it slowly once and you will answer it in one breath.

What failed, and why

The first version of the WPI Fuel and Power model used the full sample from May 1995 to March 2026. On that sample, the model failed the HAC-RESET test. ‘Fail HAC-RESET’ means: adding ŷ-squared and ŷ-cubed into the equation made a jointly significant improvement, so the straight-line ADL was missing something nonlinear.

Two economic reasons were likely causes:

- Regime mixing. The full sample lumps the pre-2010 administered pricing period together with the post-2010 market-linked period. Before 2010 the government set retail prices for long stretches, so oil shocks sometimes did not transmit at all. After 2010 oil marketing companies revised prices frequently, so transmission is much faster. Forcing one set of ADL coefficients to fit both behaviours creates a kink RESET can detect.
- COVID outliers. April through September 2020 saw unprecedented moves. Brent went negative in May 2020 on the WTI benchmark and very low on Brent; Indian demand collapsed; pricing rules were effectively suspended. Leaving those months in the sample forces the model to accommodate extreme points that the rest of the sample does not look like, which again shows up as functional-form misspecification.

What we changed

- Change 1: shorten the sample to April 2010 onward so the pre-2010 administered regime no longer contaminates the estimates.
- Change 2: drop the six months from April 2020 to September 2020, which removes the most extreme COVID outliers while keeping the rest of the post-2010 sample intact.
- Change 3: keep the exchange-rate controls and the month fixed effects unchanged. Lag orders stay ADL(12, 6), so we are not tinkering with the lag structure to force a pass.

```
# Pseudocode for the specification change
sample_full    <- dates from 1995-05 to 2026-03        # failed RESET
sample_preferred <- dates from 2010-04 to 2026-03
               minus months in { 2020-04 .. 2020-09 }  # COVID dropped

fit the same ADL(12, 6) on sample_preferred:
    dln_wpi_fp_t = α + Σ φ_i dln_wpi_fp_{t-i}
                      + Σ β+_j dln_oil_inr_pos_{t-j}
                      + Σ β-_j dln_oil_inr_neg_{t-j}
                      + exchange-rate controls
                      + month fixed effects
                      + ε_t
```

What happened after the change

Table 8.2. Before and after the WPI Fuel and Power fix.

Check	Full sample (failed)	Preferred sample (passes)
Observations N	371	186
CPT+ estimate	0.2866 ($p<0.001$)	0.5205 ($p<0.001$)
CPT− estimate	0.2677 ($p<0.001$)	0.4195 ($p<0.001$)
Asymmetry p-value	0.7832	0.1642
BG12 (serial correlation)	Pass	Pass ($p = 0.0746$)
HAC-RESET (functional form)	FAIL	Pass ($p = 0.5443$)
Recursive CUSUM (stability)	Pass	Pass ($p = 0.6999$)

Note: The preferred model has a smaller N but cleaner diagnostics. The pooled full-sample model is kept in the paper only as context so the reader can see why the change was needed.

One-sentence answer to the teacher: *'We restricted the WPI Fuel and Power sample to April 2010 onward and dropped April to September 2020 because regime mixing and COVID outliers were creating a functional-form misspecification. After the restriction, HAC-RESET has p = 0.5443, BG has p = 0.0746, and recursive CUSUM has p = 0.6999, so the model now passes the mandatory gate cleanly.'*

8.7 Coefficient-level tests: are the numbers we care about real?

Wald test on CPT+ and CPT−

- What it checks: is the cumulative pass-through statistically different from zero? And is CPT+ equal to CPT−?
- H0 for significance: CPT+ = 0 (or CPT− = 0).
- H0 for symmetry: CPT+ = CPT−.
- Rule: small p for significance = real effect. Small p for symmetry = positive and negative shocks pass through differently.
- Memory hook: a Wald test is like a ruler that measures how many standard errors away from the null our estimate is. Far away = small p.

Attenuation Wald test (the headline test of the paper)

- What it checks: is Stage 1 pass-through (Brent to retail petrol) equal to Stage 3 pass-through (rupee oil to headline CPI)?
- H0: Stage 1 CPT+ = Stage 3 CPT+.
- Result: F = 14.3499, p = 0.0002. Reject.
- Meaning: the fall in pass-through from retail petrol to headline CPI is too big to be random. Attenuation is statistically real.

8.8 Bootstrap symmetry test, in plain language

- What it checks: the same H0 as the Wald symmetry test (CPT+ = CPT−), but without relying on large-sample theory for the p-value.
- How it works: resample the residuals (in blocks, circularly) 4,999 times under the imposed null, re-compute the statistic each time, and count how often a fake statistic is as extreme as the real one. That fraction IS the bootstrap p-value.
- Why we do it: monthly samples are not infinite, so asymptotic p-values can be a little off. The bootstrap gives a second opinion that does not depend on the asymptotic distribution.
- Memory hook: bootstrap = 'shuffle the data under the null, see how lucky we would need to be to get our real estimate'.

8.9 Granger causality test, in plain language

- What it checks: if we already know past values of Y, does adding past values of X help us forecast Y better?

- H0: past values of X do not help forecast Y (X does not Granger-cause Y).
- Rule: small p = X does help = X Granger-causes Y.
- Careful: Granger causality is about prediction, NOT real causation. Two variables can Granger-cause each other and still be driven by a third cause.
- Memory hook: Granger tests 'does knowing X's past make forecasts of Y sharper?'.

8.10 Other robustness tests you should know by name

Lag-sensitivity check

- What we do: re-estimate the model with shock lag q set to 4, 5, 6, 7, 8 one by one.
- Why: to make sure the main results are not a single accident of choosing q = 6.
- In our paper: the ranking in the main table does not change when q varies in that range.

Winsorising at 1 and 99 per cent

- What we do: cap extreme values in the dependent variable at the first and ninety-ninth percentiles. A point above the 99th percentile is pulled down to the 99th percentile value; below the 1st is pulled up.
- Why: extreme shocks like COVID can distort one or two coefficients out of proportion. Capping reduces their leverage without dropping observations.
- In our paper: winsorised estimates give the same qualitative headline CPI conclusion.

COVID-exclusion check

- What we do: re-estimate with the COVID months removed from the sample.
- Why: separate test of whether COVID is driving the main estimates.
- In our paper: excluding COVID does not overturn the weak headline CPI result.

Rolling-window estimation

- What we do: fit the model on a moving window (for example, 120 months at a time) and slide the window forward month by month.
- Why: to see whether the coefficient moves smoothly or breaks sharply over time.
- In our paper: the headline CPI cumulative pass-through hovers near the central estimate without crossing into significance at conventional levels.

Brent plus exchange-rate decomposition

- What we do: instead of using the rupee oil shock as one variable, use Brent and the exchange-rate change as two separate right-hand-side variables.
- Why: to check that the main conclusion does not hinge on how the shock is bundled.
- In our paper: decomposed estimates for headline WPI are very close to the rupee-shock estimates. Conclusion does not depend on bundling.

8.11 One-page memory card: every test in one table

Table 8.3. Every test in the paper, its null, and how to read a small p-value.

Test	H0 (null)	Small p means
ADF	Unit root	Stationary (good)
PP	Unit root	Stationary (good)
KPSS	Stationary	Unit root (bad for levels)
Zivot-Andrews	Unit root (with break allowed)	Stationary with a break
Bai-Perron	No structural break	Break detected
Breusch-Godfrey (BG)	No residual autocorrelation	Leftover pattern in residuals (bad)
Breusch-Pagan (BP)	Homoskedastic residuals	Residual variance unstable
Jarque-Bera (JB)	Residuals are normal	Non-normal residuals
HAC-RESET	Functional form is correct	Missing nonlinearity (bad)
Recursive CUSUM	Coefficients are stable	Parameter drift (bad)
Wald (significance)	Coefficient = 0	Coefficient is real
Wald (symmetry)	CPT+ = CPT−	Asymmetric pass-through
Wald (attenuation)	Stage 1 CPT+ = Stage 3 CPT+	Attenuation is real
Granger	X does not help forecast Y	X Granger-causes Y
Bootstrap symmetry	CPT+ = CPT−	Asymmetric pass-through (robust check)

Five-second revision before the viva: *ADF/PP reject = stationary, KPSS reject = bad. BG/RESET/CUSUM reject = bad. Wald reject = real effect. Granger reject = predictive link. For WPI Fuel and Power we shortened to post-2010 and dropped Apr-Sep 2020 so HAC-RESET passes.*

Part 9. Results: what the numbers say, layer by layer

9.1 One table with every main number

Table 9.1. Cumulative pass-through across all five layers.

Layer	Sample	N	CPT+	p	CPT−	p	Asym. p	Verdict
PPAC retail petrol	2004-08 to 2024-12	245	0.3459	<0.001	0.1912	0.0002	0.0999	Strong direct fuel; marginal asymmetry
WPI Fuel and Power (preferred)	2010-04 to 2026-03	186	0.5205	<0.001	0.4195	<0.001	0.1642	Preferred wholesale fuel; diagnostics pass
CPI Fuel and Light (bridge)	2011-05 to 2024-12	164	0.1777	0.0021	0.1058	0.1741	0.4554	Bridge evidence; shorter sample
Headline WPI	1983-05 to 2026-03	515	0.0301	0.0240	0.0374	0.0012	0.6727	Modest but significant
Headline CPI (M1)	2004-08 to 2024-12	245	0.0213	0.1220	0.0006	0.9375	0.2408	Weak, not significant

Note: CPT+ sums the β^{+} coefficients across the ADL shock lags. CPT− sums the β^− coefficients. p-values use Newey-West HAC inference. WPI models use rupee oil shocks, retail petrol uses Brent, and CPI Fuel and Light uses PPAC petrol as the shock.

9.2 Headline WPI

The headline WPI model runs from May 1983 to March 2026, with N = 515 monthly observations. The adjusted R^2 is 0.421, which means the model explains about 42 per cent of the monthly variation in headline WPI inflation.

$CPT^{+} = 0.0301$ with $p = 0.0240$ says a one per cent positive rupee oil shock is associated with a cumulative 0.03 per cent response in headline WPI. That is small in magnitude but statistically distinguishable from zero at the 5 per cent level. $CPT^{-} = 0.0374$ with $p = 0.0012$ is the negative-shock sum, also statistically significant. The asymmetry test gives $p = 0.6727$ (HAC) and $p = 0.7461$ (bootstrap). We cannot reject that positive and negative responses are equal, so the WPI layer does not support asymmetric pass-through.

In words: *A big rise in the rupee oil price does move headline WPI, but not by much. Headline WPI is too broad for oil to dominate it.*

9.3 WPI Fuel and Power (preferred specification)

This is our preferred wholesale fuel result. The sample is April 2010 to March 2026, excluding April-September 2020 (COVID). N = 186. Adjusted R^2 is about 0.68. The diagnostics pass the gate: BG $p = 0.0746$, HAC-RESET $p = 0.5443$, recursive CUSUM $p = 0.6999$.

$CPT^{+} = 0.5205$ with $p < 0.001$. A one per cent positive rupee oil shock is associated with roughly a 0.52 per cent cumulative response in WPI Fuel and Power inflation. $CPT^{-} = 0.4195$ with $p < 0.001$. Asymmetry $p = 0.1642$, so we do not reject symmetry.

Why is this layer so much bigger than headline WPI? Because this index is close to the fuel channel. It contains petrol, diesel, LPG, electricity, and coal-adjacent items, all of which track oil costs closely. The headline WPI basket mixes this with manufactured goods and food, diluting the fuel signal.

9.4 Why we reject the older full-sample pooled Fuel and Power model

We also fit a full-sample pooled WPI Fuel and Power model over 1995-05 to 2026-03 with N = 371. It gives $CPT^{+} = 0.2866$ and $CPT^{-} = 0.2677$. Both are statistically significant. But HAC-RESET fails, so the functional form is misspecified. We report that estimate as 'context only' because the full sample mixes the administered-pricing regime, the market-linked regime, and COVID outliers. The preferred post-2010 excluding-COVID specification clears the diagnostic gate cleanly and is what we use for the wholesale fuel claim.

9.5 PPAC retail petrol

The retail petrol model runs from August 2004 to December 2024, with N = 245. The adjusted R^2 is 0.236. The shock is Brent directly, because retail petrol pricing under the post-2010 rules is anchored to international product prices.

$CPT^{+} = 0.3459$ with $p < 0.001$. A one per cent positive Brent shock is associated with a cumulative 0.35 per cent response in Delhi retail petrol over the three-month shock window. $CPT^{-} = 0.1912$ with $p = 0.0002$. Asymmetry $p = 0.0999$.

Asymmetry reading: *The asymmetry p-value just below 0.10 is the one place in the paper where asymmetry is suggestive. It is the well-known 'rockets and feathers' pattern: retail petrol rises faster than it falls. But we are careful to say 'marginal at the 10 per cent level', not 'significant asymmetry'.*

9.6 CPI Fuel and Light bridge

This equation uses PPAC petrol as the shock and CPI Fuel and Light as the dependent. The sample is May 2011 to December 2024, N = 164. The shorter sample is why we call it bridge evidence.

$CPT^{+} = 0.1777$ with $p = 0.0021$. A one per cent positive retail petrol shock is associated with a cumulative 0.18 per cent response in CPI Fuel and Light. $CPT^{-} = 0.1058$ with $p = 0.1741$ is not significant. Asymmetry $p = 0.4554$.

The bridge shows that retail fuel movements do enter the fuel-sensitive part of the consumer basket, at a size smaller than retail petrol but larger than headline CPI. That is exactly the shape attenuation would produce.

9.7 Headline CPI endpoint

The headline CPI model uses rupee oil as the shock. Sample 2004-08 to 2024-12, N = 245, adjusted $R^2 = 0.4492$. Diagnostics pass the mandatory gate.

$CPT^{+} = 0.0213$ with $p = 0.1220$. The positive-shock cumulative response has the expected sign but is NOT statistically significant at conventional levels. $CPT^{-} = 0.0006$, $p = 0.9375$, is essentially zero. Asymmetry p = 0.2408 does not reject symmetry. Bootstrap symmetry p = 0.4997.

Key takeaway: *This is the crucial number. Headline CPI is where oil becomes statistically quiet. We describe it as weak and not significant, NEVER as 'zero' or 'no effect'.*

9.8 The integrated attenuation Wald test

To make the attenuation story formal we estimate the consumer-side chain on a common sample window and then test whether the first-stage pass-through equals the third-stage pass-through.

Table 9.2. Consumer-side chain, common-sample estimates.

Stage	What it measures	CPT+	p
Stage 1	Brent → PPAC retail petrol	0.4007	0.0001
Stage 2	PPAC petrol → CPI Fuel and Light	0.1777	0.0021
Stage 3	Rupee oil → Headline CPI	0.0064	0.6355

Formal Wald test: H_0: $CPT^{+}_{Stage\ 1} = CPT^{+}_{Stage\ 3}$. F = 14.3499, p = 0.0002. We reject equality. Attenuation across the consumer-price chain is statistically supported.

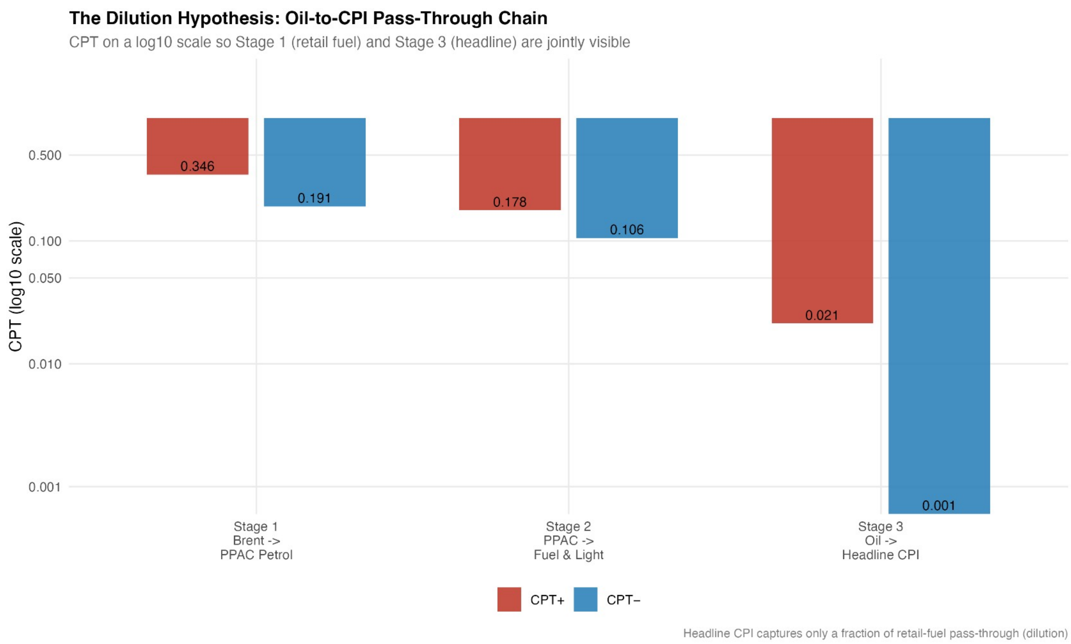

Figure 9.1. Cumulative positive pass-through across the consumer-price chain, showing attenuation from retail petrol to headline CPI.

9.9 Pre/post-2010 wholesale split (institutional context)

Table 9.3. Pre/post-2010 wholesale pass-through split.

Model	Pre-2010 CPT+ (p)	Post-2010 CPT+ (p)
Headline WPI	0.0117 (p = 0.3812)	0.0741 (p = 0.0043)
WPI Fuel and Power	0.0922 (p = 0.1679)	0.5241 (p < 0.001)

Note: The split is institutional evidence consistent with stronger pass-through under market-linked fuel pricing. It is NOT a clean causal estimate of deregulation.

Both rows show the post-2010 estimates are materially larger and statistically stronger. Under more market-linked pricing, a given oil shock passes through to wholesale more of the time and faster. Other things changed over the same period too: new CPI series 2011, flexible inflation targeting 2016, GST 2017, and several fuel-tax episodes. So we do not call this a clean deregulation experiment.

9.10 Predictive precedence (Granger tests)

- Rupee oil → headline WPI: $F = 7.43$, $p < 0.001$. Supported.
- Rupee oil → WPI Fuel and Power: $F = 15.22$, $p < 0.001$. Supported.
- Brent → PPAC petrol: $F = 10.71$, $p < 0.001$. Supported.
- Rupee oil → headline CPI: $F = 2.29$, $p = 0.0794$. Only at the 10 per cent level. Weak.
- PPAC petrol → CPI Fuel and Light: $p = 0.113$. Not supported.

This lines up with the main estimates: the oil signal is easy to pick up near fuel prices and harder to detect in the broad CPI.

9.11 The one-line summary of the results chapter

Remember this: *Strong in retail petrol and WPI Fuel and Power. Medium in CPI Fuel and Light. Modest in headline WPI. Weak and not significant in headline CPI. Wald test formally rejects that retail petrol and headline CPI positive pass-throughs are equal.*

Part 10. Robustness checks and limitations

10.1 Six robustness checks, what each one shows

- Brent plus exchange-rate decomposition: we split rupee oil into its two drivers and re-estimate headline WPI. $CPT^{+} = 0.0309$ ($p = 0.0234$), $CPT^{-} = 0.0375$ ($p < 0.001$), asymmetry $p = 0.7039$. Nearly identical to the rupee-shock result. Conclusion: the headline WPI result does not depend on how we bundle the two components.
- Bootstrap symmetry: 4,999 restricted-residual circular block replications. Symmetry is not rejected for headline WPI ($p = 0.7461$), the full-sample WPI Fuel and Power context model ($p = 0.8196$), and headline CPI ($p = 0.4997$). Asymmetry is not the headline finding.
- Granger predictive tests: direction from oil to wholesale is supported; reverse direction is not. Matches the layered reading.
- Bai-Perron break tests: where breaks show up in the WPI inflation series, the institutional pre/post-2010 split is the policy-relevant way of handling them.
- COVID window and winsorising: excluding COVID from the headline CPI specification, and winsorising at the 1st and 99th percentiles, leaves the main CPI conclusion unchanged.
- Lag-sensitivity: varying the shock lag q between 4 and 8 in the wholesale models does not change the ranking in Table 9.1.

10.2 Table of diagnostic pass/fail status at a glance

Table 10.1. Diagnostic and symmetry status for every claim-bearing or context specification.

Spec	BG12	HAC-RESET	Rec-CUSUM	Bootstrap sym. p
Headline WPI	Pass	Pass	Pass	0.7461
WPI F&P preferred (post-2010, excl. COVID)	Pass	Pass	Pass	Asym. HAC p = 0.1642
WPI F&P pooled context	Pass	Fail	Pass	0.8196
Headline CPI M1	Pass	Pass	Pass	0.4997
PPAC retail petrol	Pass	Pass	Pass	HAC asym. p = 0.0999

Note: BG12 and HAC-RESET are described in Part 8. The bootstrap symmetry p-value is reported where computed; otherwise the HAC-based asymmetry p-value is shown for completeness.

10.3 Limitations, stated plainly

- Reduced-form models, not structural causal estimates. We project; we do not identify a structural shock.
- Different layers have different sample windows, because the official publishers made each series available at different dates. The common-sample CPI chain is the check that handles this concern.
- CPI Fuel and Light starts only in 2011, so it is bridge evidence, not a 20-year headline claim.

- PPAC retail petrol uses Delhi prices as a retail-fuel proxy. Other cities have similar dynamics, but we only run the estimate on Delhi.
- The full-sample pooled WPI Fuel and Power model fails the functional-form test. We report it as context and move the wholesale fuel claim to the post-2010 excluding-COVID model.
- The pre/post-2010 wholesale split is institutional evidence, NOT a clean causal estimate of deregulation.
- Headline CPI pass-through should be described as weak and not statistically significant, NOT as 'zero'.
- The layered map is not a single mechanical shock moving through five equations. It is five related reduced-form equations whose coefficients share a common interpretation.

What to say if a reviewer presses on causality: *'Our estimates are reduced-form short-run projections. The Granger tests show predictive precedence from oil to wholesale and to retail fuel, but we do not make structural causal claims. For structural identification, a shock-based SVAR or a natural experiment would be needed; that is a logical future extension.'*

Part 11. Conclusion and the policy takeaways

11.1 Direct answer to the research question

Oil-price shocks transmit to India's prices in a layered way. They are strong in retail petrol (CPT+ = 0.3459) and in the preferred WPI Fuel and Power model (CPT+ = 0.5205). They are still visible in CPI Fuel and Light (CPT+ = 0.1777). They are modest but statistically significant in headline WPI (CPT+ = 0.0301). And they are weak and not statistically significant in headline CPI (CPT+ = 0.0213). The formal Wald test rejects equality between the retail petrol and headline CPI positive pass-through ($F = 14.3499$, $p = 0.0002$). So the pass-through weakens specifically between the fuel layers and the broad consumer index.

11.2 Policy takeaways, carefully stated

- A monetary-policy framework that relies only on headline CPI may miss oil-side pressure, because oil's footprint at that layer is statistically small.
- Monitoring a small set of fuel-sensitive layers (retail petrol, WPI Fuel and Power, CPI Fuel and Light) gives more short-run information about oil pass-through than the headline alone.
- The larger post-2010 wholesale pass-through is consistent with a more market-linked fuel pricing regime. It is not a clean causal estimate of deregulation, so the wording should stay cautious.
- Asymmetric pricing (rockets and feathers) is marginal at retail petrol, and not supported at the headline layers.

11.3 What we do not claim

- We do not claim oil is irrelevant for Indian inflation.
- We do not claim deregulation caused bigger pass-through.
- We do not claim asymmetric pass-through is the headline finding.
- We do not claim a single structural chain from Brent to headline CPI.
- We do not claim headline CPI pass-through is zero. It is weak and not significant in our sample.

11.4 Natural extensions for future work

- Decompose headline CPI into transport, housing, and miscellaneous services and trace where the small headline signal sits.
- Fit a time-varying-parameter ADL so the coefficients can drift smoothly with the pricing regime rather than breaking at a single date.
- A structural VAR with identified oil-supply and exchange-rate shocks, to convert the reduced-form map into a causal decomposition.
- A cross-country comparison of attenuation maps across oil-importing emerging economies.

11.5 Closing line, the one the teacher will remember

Oil shocks do not vanish in India, but they lose force as they move from fuel prices to broad consumer inflation.

If you only remember one thing: *This study turns oil pass-through into a layered attenuation map. The numbers fall cleanly from retail petrol and WPI Fuel and Power down to headline CPI, and the formal Wald test says that fall is statistically real.*

Part 12. Viva question bank with short model answers

12.1 Setting-the-scene questions

Q. What is your main research question?

A. How global oil-price shocks transmit across India's wholesale, retail fuel, fuel-sensitive consumer, and headline consumer price layers, and where that pass-through weakens.

Q. What is your main finding in one line?

A. Oil-price shocks pass through strongly at fuel-related layers and weakly at the headline CPI layer, and the difference is statistically significant.

Q. What is the contribution of the study?

A. Treating oil pass-through as a layered transmission problem across five price indices, rather than as a single elasticity between Brent and headline CPI, and backing that framing with a formal Wald test on the common-sample consumer-price chain.

Q. What is novel compared with earlier India studies?

A. Earlier work usually picks one index or one equation. We put five related equations into one attenuation map, keep each layer's sample and shock variable visible, and show that the fall in pass-through from retail petrol to headline CPI is statistically real.

Q. Why does this matter for policy?

A. The Reserve Bank of India targets headline CPI for inflation. If headline CPI is weakly tied to oil, a policy reader looking only at headline might miss upstream fuel-side pressure. So they should also watch fuel-sensitive layers.

12.2 Data questions

Q. Where does your Brent crude data come from?

A. The World Bank Commodity Price Data, commonly called the Pink Sheet. Monthly frequency. Cross-checked against FRED POILBREUSDM.

Q. Where does the INR/USD exchange rate come from?

A. FRED series EXINUS for the CPI pipeline, and EXINUS_latest for the WPI pipeline. FRED is Federal Reserve Economic Data at the Federal Reserve Bank of St. Louis.

Q. Who publishes the WPI?

A. The Office of the Economic Adviser (OEA), inside the Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry.

Q. Who publishes the CPI?

A. The Ministry of Statistics and Programme Implementation, MoSPI.

Q. Why PPAC retail petrol for Delhi and not national?

A. PPAC (Petroleum Planning and Analysis Cell, Ministry of Petroleum and Natural Gas) publishes consistent monthly Delhi prices back to 2004. It is used as a retail-fuel channel proxy. Other metros give similar qualitative patterns.

Q. What is the rupee oil price? Why construct it?

A. It is Brent in dollars multiplied by the INR/USD rate. We use it because Indian refiners and the broader economy pay for oil in rupees. A Brent rise with a rupee appreciation at the same time creates a different domestic cost pressure than a Brent rise alone.

Q. Why do your samples differ across layers?

A. Each official series becomes available at different dates. Headline WPI goes back to 1983-05, WPI Fuel and Power to 1995-05, retail petrol to 2004-08, and CPI Fuel and Light only to 2011-05. We use the longest consistent window for each layer.

Q. How do you handle the fact that samples differ?

A. We run a common-sample consumer-chain check where all three stages are estimated on overlapping months, with small observation differences from lag construction. The qualitative ranking holds. The Wald test for attenuation uses this common-sample set-up.

12.3 Variable questions

Q. Why log differences and not levels?

A. Price levels have unit roots and trend together. Regressing them on each other produces spurious results. Log differences are stationary and also read naturally as percentage changes.

Q. Why multiply the log difference by 100?

A. So that each coefficient reads as a percentage response per one per cent shock. Without the 100 the economics is the same, just in decimals.

Q. What is the positive/negative shock split?

A. For every shock we build a non-negative ‘twin’ that equals $\max(\Delta x, 0)$ and a non-positive ‘twin’ that equals $\min(\Delta x, 0)$. They sum to the original shock. This lets us test whether positive and negative shocks pass through at the same size.

Q. What is CPT+ and CPT−?

A. CPT+ is the sum of coefficients on the positive-shock lags, adding up the total cumulative response to a one per cent positive shock over the ADL lag window. CPT− is the same sum for negative shocks.

12.4 Methodology questions

Q. Why ADL?

A. ADL models allow the dependent inflation to depend on its own recent past (autoregressive) and on current and recent shocks (distributed lag). That matches how price data behave. We use the asymmetric version to test rockets and feathers.

Q. Why not NARDL or an error-correction model?

A. Our question is about monthly transmission, not about long-run equilibrium relationships. NARDL bounds and error-correction tools would be overkill and would invite long-run claims we do not need.

Q. What does ADL(3,3) mean? How did you choose 3 and 3?

A. Three lags of the dependent variable, three lags of the shock. The dependent-lag order is picked by AIC over $p = 1$ to 4 for the CPI and retail petrol pipelines. The shock-lag order balances fast retail and consumer dynamics against the informational content of older lags.

Q. Why ADL(12,6) for WPI?

A. Twelve own-lags capture the annual cycle in wholesale inflation even after month dummies. Six shock-lags let the oil response stretch half a year, which matches the way wholesale prices move after input costs change.

Q. How are the coefficients estimated? Is it OLS or something fancier?

A. Plain OLS. We estimate every ADL equation by minimising the sum of squared residuals. The only non-standard step is the standard errors, which are Newey-West HAC to fix autocorrelation and heteroskedasticity.

Q. What do α, β, γ, φ, μ, ε mean in your equation?

A. α is the intercept. φ_i are coefficients on own-lags of the dependent variable. β^+_j and β^-_j are coefficients on positive and negative shock lags. γ is the vector of coefficients on controls. μ_m are month-of-year dummies. ε_t is the residual, everything the model did not explain.

Q. Are there specific numerical values of these coefficients?

A. Yes; the full coefficient tables are in the model output folders. In the main text we report the cumulative pass-through CPT+ and CPT− because those are the quantities policy readers care about.

12.5 Inference questions

Q. What are Newey-West HAC standard errors?

A. Standard errors that remain valid when residuals are heteroskedastic (different variance over time) and autocorrelated (residuals correlated with their own recent past). Without HAC, p-values would be too small and we would over-reject the null.

Q. Why 4,999 bootstrap replications?

A. Odd numbers avoid ties when computing percentile p-values. 4,999 is enough for stable estimates. Using 9,999 would move the p-values by a small amount at most.

Q. What does a Wald test do?

A. It tests linear restrictions on coefficients. Our CPT+ = 0 and CPT+ = CPT− tests are Wald tests. Under HAC inference, a large Wald statistic gives a small p-value and rejects the null.

Q. How do Granger tests fit into the paper?

A. They are predictive-precedence tests, not causality tests. They show that past oil movements improve forecasts of wholesale and retail fuel inflation. We report them as supporting evidence, not as structural causal identification.

12.6 Diagnostic questions

Q. What is the mandatory model gate?

A. Three tests every claim-bearing model must pass: Breusch-Godfrey at 12 lags for serial correlation, HAC-RESET for functional form, and recursive CUSUM for parameter stability.

Q. Why did the full-sample WPI Fuel and Power model fail?

A. It fails HAC-RESET, meaning the functional form is misspecified. The most likely reason is regime mixing: it combines the pre-2010 administered period with the post-2010 market-linked period and the COVID outliers. Restricting to post-2010 and dropping April-September 2020 fixes this.

Q. What do ADF, PP, KPSS do?

A. They test for unit roots. ADF and PP have H_0: unit root. KPSS reverses it with H_0: stationarity. Using all three together gives a robust verdict. All our differenced series pass as stationary.

Q. What about the KPSS caveat on WPI inflation?

A. KPSS rejects stationarity for WPI inflation in a couple of subsamples. We read this alongside the Bai-Perron break test results and treat it as a break-related caveat handled by the institutional pre/post-2010 split, not a reason to re-difference.

12.7 Results questions

Q. Why is CPT+ so much larger in WPI Fuel and Power than in headline WPI?

A. WPI Fuel and Power is close to the fuel channel. Its basket contains petrol, diesel, LPG, coal-adjacent items, and electricity. Headline WPI averages this with manufactured goods and food, so the oil signal is diluted.

Q. Why is headline CPI pass-through weak?

A. The CPI basket is dominated by food, services, housing, health, and education. Fuel and Light is just one group. So even a big shock in fuel prices becomes a small number by the time you sum to headline CPI.

Q. Does headline CPI pass-through equal zero?

A. No. It is weak and statistically insignificant at conventional levels in our sample. We never describe it as zero. We describe it as weak and not significant.

Q. What does the Wald F = 14.35 mean in words?

A. It is the test statistic for the null that Stage 1 CPT+ equals Stage 3 CPT+ in the common-sample consumer-price chain. With $p = 0.0002$, we reject the null. The attenuation from retail petrol to headline CPI is statistically real.

Q. Why is asymmetry not the headline finding?

A. Because HAC and bootstrap symmetry tests fail to reject symmetry in headline WPI, WPI Fuel and Power, CPI Fuel and Light, and headline CPI. Retail petrol is the only layer with even marginal evidence, at $p = 0.0999$.

Q. Why did you prefer the post-2010 Fuel and Power model?

A. It is the only Fuel and Power specification that passes the mandatory diagnostic gate cleanly. The full-sample pooled model fails HAC-RESET because it mixes regimes.

12.8 Broader and trickier questions

Q. Is your work causal?

A. No. It is reduced-form projection. Granger tests show predictive precedence from oil to wholesale and retail fuel, but we do not claim structural identification.

Q. Could the pre/post-2010 split be confounded?

A. Yes. The post-2010 window also contains the new CPI series in 2011, flexible inflation targeting in 2016, GST in 2017, and several fuel-tax episodes. We call the split institutional evidence, not causal.

Q. Why Delhi for retail petrol?

A. PPAC has the longest consistent monthly Delhi series. Other metros behave similarly. We use Delhi as a retail-fuel channel proxy.

Q. Could endogeneity be a problem? Does India affect global Brent?

A. India is a large buyer but not a price-setter for Brent. Brent is driven by OPEC+ decisions, US shale, global demand, and geopolitics, all exogenous from India's perspective. The reverse-direction Granger tests also fail to reject the null that wholesale Indian prices do not predict Brent. So endogeneity concerns are small.

Q. What if oil's effect operates through expectations rather than cost?

A. That is possible and is one of the reasons headline CPI is not the layer where oil transmission is most visible. Expectations feed into the whole basket. Our layered evidence is consistent with direct cost transmission being stronger at fuel layers; expectation channels would be a structural-shock extension.

Q. Would a structural VAR improve the paper?

A. It would allow identified oil-supply, demand, and exchange-rate shocks, and shock-specific impulse responses. That is a legitimate extension. We chose ADL because the question is short-run pass-through by layer, and ADL keeps the interpretation clean.

Q. How would your results change if CPI were defined differently?

A. We use the standard MoSPI headline CPI and its Fuel and Light component. Alternative definitions (core CPI, rural-only, urban-only) would move the headline number by small amounts. The attenuation pattern is robust because it depends on basket breadth, not on the exact CPI variant.

Exam tip: *When in doubt, bring the answer back to layered attenuation. Whether the teacher asks about method, data, or a specific number, the story is: oil is strong in fuel layers, weak in headline CPI, and the Wald test formally confirms the fall.*

Part 13. Glossary and cheat sheet

13.1 Glossary of acronyms and terms

Table 13.1. Acronyms and terms used throughout the paper and this guide.

Term	Full form / meaning
ADL	Autoregressive Distributed Lag model
ADF	Augmented Dickey-Fuller unit-root test
AIC	Akaike Information Criterion
Attenuation	A reduction in the size of a shock's effect as it moves through a chain
Bai-Perron	Structural-break test for multiple unknown breaks
BG	Breusch-Godfrey serial correlation test
BIC	Bayesian Information Criterion
Brent	North Sea crude benchmark used as the global oil price reference
CPI	Consumer Price Index (MoSPI)
CPI Fuel and Light	CPI sub-group covering household energy
CPT+	Cumulative positive pass-through
CPT−	Cumulative negative pass-through
CUSUM	Cumulative sum parameter-stability test
DPIIT	Department for Promotion of Industry and Internal Trade, Ministry of Commerce and Industry
EXINUS	FRED INR/USD exchange rate series
FRED	Federal Reserve Economic Data, St. Louis Fed
GST	Goods and Services Tax (India, from July 2017)
HAC	Heteroskedasticity and Autocorrelation Consistent standard errors (Newey-West)
HQIC	Hannan-Quinn Information Criterion
IIP	Index of Industrial Production (MoSPI)
INR	Indian Rupee
KPSS	Kwiatkowski-Phillips-Schmidt-Shin stationarity test
MoCI	Ministry of Commerce and Industry
MoPNG	Ministry of Petroleum and Natural Gas
MoSPI	Ministry of Statistics and Programme Implementation
NIPFP	National Institute of Public Finance and Policy
OEA	Office of the Economic Adviser, DPIIT, MoCI
OLS	Ordinary Least Squares

Term	Full form / meaning
Pink Sheet	World Bank's monthly commodity price data bulletin
PP	Phillips-Perron unit-root test
PPAC	Petroleum Planning and Analysis Cell, MoPNG
RESET	Ramsey Regression Equation Specification Error Test
USD	United States Dollar
Wald test	Test of linear restrictions on coefficients
WPI	Wholesale Price Index (Office of the Economic Adviser)
WPI Fuel and Power	WPI sub-group covering fuel, power, and mineral oils
Zivot-Andrews	Unit-root test allowing one structural break

13.2 One-page cheat sheet you can glance at during the presentation

- Question: where does oil pass-through weaken across Indian price layers?
- Answer: it weakens between fuel layers and the broad headline consumer index.
- Main result numbers: CPT+ = 0.5205 preferred WPI F&P ($p<0.001$); 0.3459 retail petrol ($p<0.001$); 0.1777 CPI F&L bridge ($p=0.0021$); 0.0301 headline WPI ($p=0.024$); 0.0213 headline CPI ($p=0.122$, not significant).
- Formal attenuation test: $F = 14.3499$, $p = 0.0002$. Reject equality between retail petrol and headline CPI CPT+.
- Asymmetry: only marginal at retail petrol ($p=0.0999$). Not the headline finding.
- Method: short-run asymmetric ADL in monthly log differences, OLS with Newey-West HAC standard errors, bootstrap symmetry check with 4,999 replications.
- Lag orders: ADL(12,6) for WPI models, ADL(3,3) for CPI and retail petrol models.
- Diagnostic gate: Breusch-Godfrey, HAC-RESET, recursive CUSUM. All claim-bearing models pass.
- Data sources (full form): OEA WPI, MoSPI CPI, PPAC Delhi petrol, World Bank Pink Sheet Brent, FRED EXINUS for INR/USD.
- Core message: oil still matters in India, but most of it is absorbed before it reaches headline CPI.

Closing line to rehearse: *'Oil shocks do not vanish in India, but they lose force as they move from fuel prices to broad consumer inflation.'*